

1. EDR vs XDR

Feature	EDR (Endpoint Detection & Response)	XDR (Extended Detection & Response)
Focus	Endpoints only (laptops, servers, workstations)	Multiple security layers
Data Sources	Endpoint telemetry	Endpoints, network, email, cloud, identity, servers
Visibility	Limited to endpoint activities	Organization-wide visibility
Threat Detection	Detects endpoint attacks	Correlates threats across multiple systems
Investigation	Endpoint-centric	Cross-platform investigation
Example	Malware on a laptop	Phishing email → credential theft → cloud access → data exfiltration

Examples:

- EDR: Microsoft Defender for Endpoint, CrowdStrike Falcon.
- XDR: Microsoft Defender XDR, Palo Alto Cortex XDR, Trend Micro Vision One.

2. Attack Scenario

Scenario: Phishing Attack Leading to Ransomware

1. Attacker sends a phishing email to an employee.
2. Employee clicks the malicious attachment.
3. Malware installs on the workstation.
4. Malware steals credentials.

5. Attacker moves laterally across the network.
6. Sensitive files are accessed and encrypted.
7. Ransom note is displayed demanding payment.

What EDR Sees

- Suspicious process execution.
- Malware installation.
- File encryption behavior.

What XDR Sees

- Phishing email delivery.
- Endpoint compromise.
- Credential theft.
- Lateral movement.
- Unusual network traffic.
- Data exfiltration attempts.

3. Suggested Response

Immediate Actions

1. Isolate the affected endpoint from the network.
2. Disable compromised user accounts.
3. Block malicious IPs, domains, and hashes.
4. Terminate malicious processes.
5. Preserve logs and forensic evidence.

Investigation

1. Identify patient zero.
2. Determine attack timeline.
3. Check whether data was exfiltrated.
4. Review all affected systems.
5. Search for Indicators of Compromise (IOCs).

Recovery

1. Remove malware.
2. Patch vulnerabilities.
3. Reset passwords and enforce MFA.
4. Restore systems from clean backups.
5. Monitor for reinfection.

Prevention

- Security awareness training.
- Email filtering.
- Multi-factor authentication.
- EDR/XDR deployment.
- Least privilege access.
- Regular vulnerability management.